

Boosting Visual Fidelity in Driving Simulations through Diffusion Models

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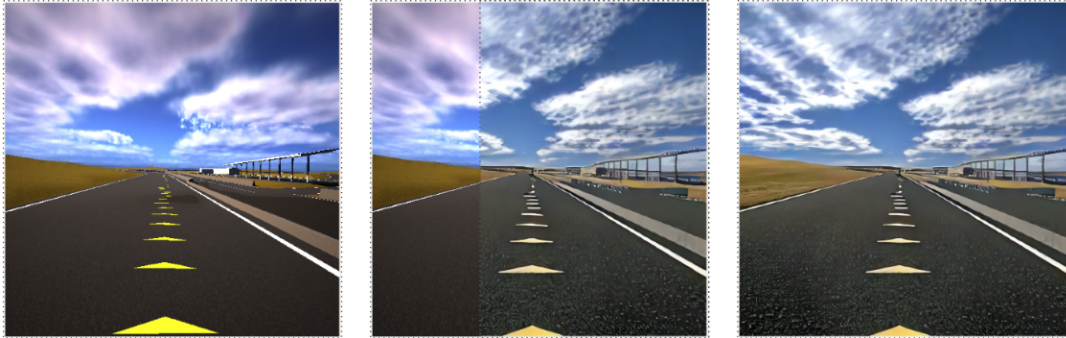


Figure 1: Using diffusion models, we style transfer the view within a driving simulation into a photorealistic image in near real-time. The leftmost image is taken within a driving simulation in Unreal Engine, and the rightmost image is the output of our diffusion pipeline.

Abstract

Diffusion models have made substantial progress in facilitating image generation and editing. As the technology matures, we see its potential in the context of driving simulations to enhance the simulated experience. In this paper, we explore this potential through the introduction of a novel system designed to boost visual fidelity. Our system, DRIVE (Diffusion-based Realism Improvement for Virtual Environments), leverages a diffusion model pipeline to give a simulated environment a photorealistic view, with the flexibility to be adapted for other applications. We conducted a preliminary user study to assess the system’s effectiveness in rendering realistic visuals and supporting participants in performing driving tasks. Our work lays the groundwork for future research on the integration of diffusion models in driving simulations, and provides practical guidelines and best practices for their application in this context.

CCS Concepts

• **Computing methodologies** → **Simulation environments**; • **Human-centered computing** → *Usability testing*.

Keywords

Diffusion Models, Driving Simulation, Visual Fidelity

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1 Introduction

Driving simulators have been widely used in both academic research and industry due to their cost-effectiveness and safety benefits [1, 18]. As driving simulations serve as proxies for real-world driving environments, the question of validity is a recurring topic of discussion. While various definitions of validity exist in the context of driving simulations, physical validity and behavioral validity are the two commonly accepted forms of validity to evaluate the effectiveness of driving simulations [10]. Physical validity reflects how the simulator replicates its real-world counterpart, including aspects such as visual fidelity and vehicle dynamics; behavioral validity concerns whether the driving simulator can enable participants to behave as if the simulated situation were real. In our work, we investigate the improvement of physical validity through visual enhancement. Research has shown that visual fidelity can significantly influence drivers’ performance and behavior, particularly in attention-critical applications, as varying levels of visual fidelity may alter drivers’ attentional responses [28, 39, 49]. Consequently, our work aims to enhance the visual fidelity of existing driving simulations, hoping such improvement will lead to increased behavioral validity in downstream user studies.

Traditional methods to increase visual fidelity in driving simulations involve powerful rendering engines and labor-intensive efforts

to create realistic textures and meshes [5, 9, 29, 50]. Such efforts typically require multiple expert artists and are challenging to generalize across different simulated environments. With advancements in computer graphics technologies, particularly generative models, there is a growing shift from rendering-based methods in 3D space to data-driven approaches in 2D-pixel space [3, 19, 22, 34, 46], offering better scalability.

In this paper, we explore the use of diffusion models as a data-driven method to improve visual fidelity directly in the image space. Building on previous research, we minimize the rendering effort in the 3D scene and delegate the task of aesthetic enhancement to a data-driven diffusion model. Consequently, we introduce the DRIVE system, an interactive platform that shifts the burden of high-fidelity rendering from the simulation engine to a diffusion model. We demonstrate the capabilities of the DRIVE system through a preliminary user study and discuss its potential for other forms of visual enhancement as well.

2 Related Works

2.1 Photorealistic Driving Simulations

The traditional approach to achieving photorealistic driving simulations relies on powerful rendering engines, as well as meticulously crafted meshes and textures [5, 9, 35]. However, new 3D models and textures must be created when adapting to new environments. For example, in GarchingSim, Zhou et al. [50] employs a team of specialized artists to create assets in the Unity3D game engine. To achieve a photorealistic scene in Unity, they also have to implement the High Definition Render Pipeline (HDRP) in Unity.

To reduce costly human labor, hybrid methods that combine image-based and geometry-based techniques have been proposed. Ono et al. [31] designed a system where only nearby objects, typically those users interact with, are rendered using geometry-based methods, while distant objects are rendered from a recorded video image dataset using image-based techniques. With advancements in generative models, the image-based component of this hybrid approach can be replaced with synthesized backgrounds generated by Generative Adversarial Networks (GANs) [46], and the generation process is usually conditioned with a semantic map to impose constraints [34, 46]. The synthesis of foreground-rendered objects with background images is typically achieved through dedicated hardware or blending techniques like alpha blending.

Leveraging internet-scale data, researchers can now simulate dynamic environments purely in pixel space without a backend physics engine. Neural radiance fields (NeRFs) have gained popularity in separately modeling the foreground dynamic objects and background static environments with real-world image/video datasets, enabling the simulation of novel, photorealistic vehicle behavior or changing camera view angles [38, 41, 44]. To address the limitation of NeRFs' slow training and rendering process, Gaussian Splatting techniques are suggested as an alternative for real-time rendering [4, 43]. Moreover, the data-driven neural simulator proposed by Kim et al. [19] is trained on recorded driving videos and associated human maneuvers to learn the transition between frames. As such, the simulator can output the next frame purely based on the previous frame and input action, without the help of traditional physics engines.

With the help of machine learning and large datasets, we are witnessing a shift from complex 3D rendering to sophisticated 2D image post-processing in the literature of driving simulations.

2.2 Image Style Transfer

Once in image space, the task of transforming a 2D image captured in simulation into a photorealistic scene can be approached as a style transfer problem [37, 45]. This involves transferring the content of the input image to match the style of a realistic image. Researchers have applied style transfer algorithms in various contexts, from enhancing visual fidelity in surgical training to improving sim-to-real transfer in robot learning [16, 25]. With advancements in generative models, the primary methods for style transfer have evolved from traditional image feature matching to data-driven approaches, such as diffusion models [13, 23, 40, 42, 48]. There are multiple techniques for performing style transfer using diffusion models, with some of the simplest methods involving prompt engineering or fine-tuning existing models like Stable Diffusion [8]. Style transfer using diffusion models is an active area of research and extends beyond the scope of this paper. In our approach, we employed a parameter-efficient fine-tuning method, Low-Rank Adaptation (LoRa) [15], as a proof of concept.

3 The DRIVE System

In this section, we introduce the DRIVE system, which harnesses the power of diffusion models in driving simulations (Fig. 2). To ensure the system's generalizability and adaptability across various applications, we used Robot Operating System 2 (ROS 2 Humble¹) to facilitate the communications among different components within the system. A system diagram is shown in Fig. 2. The input raw image data can originate from diverse sources, including pre-existing Sensor Simulators like Carla [5], 3D Game Engines like Unreal Engine and Unity [7, 12], or screen Capture devices. ROS 2 will fetch images from different modules within the system and broadcast them back as Image type messages. While the system supports diverse sources, the following sections describe the system with Unreal Engine 5.3 as an example.

3.1 Pipeline Overview

The core of the DRIVE system is a diffusion model pipeline designed to process image streams in real time. Our pipeline leverages the existing Stable Diffusion V1.5 framework² [33]. To optimize computational efficiency and accelerate inference, we have substituted the original variational autoencoder with a Tiny Autoencoder (TAESD)³, a modification commonly adopted within the community. The architecture of our pipeline is illustrated in Figure 4. The remainder of this section delves into the essential requirements for deploying the diffusion model in driving simulations and discusses our approach to ensuring the model adheres to these constraints.

¹<https://docs.ros.org/en/humble/index.html>

²<https://huggingface.co/runwayml/stable-diffusion-v1-5>

³<https://github.com/madebyollin/taesd>

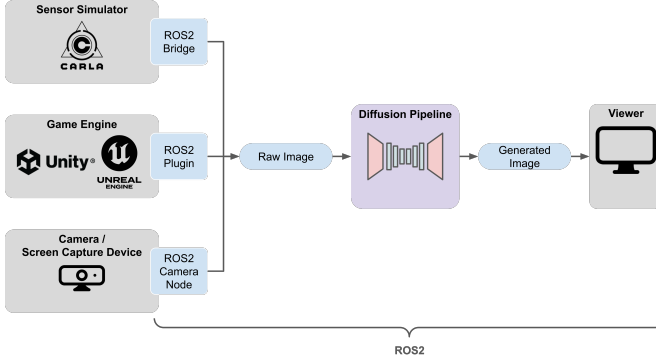


Figure 2: A system diagram for the DRIVE system.

3.2 Acceleration

The diffusion pipeline must process images in real time to maintain the basic functionality of driving simulations. To this end, we combined multiple approaches to accelerate our pipeline.

We integrated distilled latent consistency models (LCM) in our pipeline to reduce the number of required inference steps per image [26]. Specifically, we injected LCM LoRa adapters in our pipeline, adjusting the pipeline to process images at a resolution of 640 x 480 for only one inference step [27]. The model is further accelerated by the Stable Fast acceleration optimizer⁴. Lastly, we used two different machines, one for Unreal Engine and one for the diffusion pipeline, to distribute the computation load. Currently, Unreal Engine operates on a laptop equipped with an i9-9900KF CPU and an Nvidia GeForce RTX 2080 Mobile GPU. Meanwhile, the diffusion model runs on a desktop powered by an i9-11900K CPU and an Nvidia GeForce RTX 3090 GPU. Note that since the aesthetic is now handled by the diffusion pipeline, the 3D rendering in Unreal Engine can be kept as simple as possible; users can reduce rendering quality in the simulation to save computing without affecting the output quality from the diffusion pipeline. The communication between computers is also managed by ROS 2. This division of computational tasks ensures each component of our system is running on dedicated hardware that maximizes its performance potential.

Currently, our system achieves a processing rate of 10 fps for an input resolution of 640 x 480, demonstrating its potential as an experimental framework for real-time driving simulation.

3.3 Within-frame Consistency

We define *within-frame consistency* as the requirement to keep a selective set of features from the input image invariant in the corresponding output image. In the context of driving, we must guarantee the shape and contour of lane markings on the road remain the same in the output image, which is critical for driving tasks. To address this concern, we leverage ControlNet to condition the generative process with an edge map of the input image (Fig. 4) [47]. Through ControlNet, we inject spatial conditioning controls into the pre-trained model to maintain the contours invariant between every input image and its associated output image.

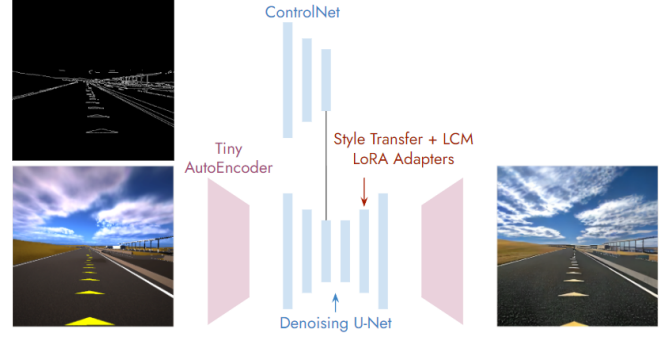


Figure 3: Diffusion model pipeline that gives our simulation a photorealistic view.

3.4 Cross-frame Stability

In the context of driving simulations, the information contained within consecutive frames is highly overlapped. We define *cross-frame stability* as the necessity to maintain the overlapping information consistent throughout the frames. For example, if a vehicle is approaching a landmark, the landmark must have the same appearance in all images that capture it. While maintaining such consistency is essential in all driving simulations, it poses a unique challenge for diffusion models, which are inherently generative. Even if the same image is input multiple times, diffusion models may produce vastly different outputs. In a way, the diffusion model behaves like a one-to-many function; however, our goal is to achieve a one-to-one mapping, where the same input reliably produces the same output.

The output variability is primarily due to the noise introduced during the diffusion process. To counter this, we add an identical noise pattern to each image by re-seeding the pipeline with the same seed for every image, thereby eliminating the stochastic nature of the diffusion process and imposing deterministic behavior on the model.

3.5 Simulation Environment and Photorealistic Style Transfer

We explore a specific application of our diffusion pipeline aimed at enhancing the realism of driving simulations. Implemented in Unreal Engine (UE) 5.3, our simulation environment features a simplified Thunderhill Race Track located in Willows, California, and users can drive and explore the virtual environment freely.

To give the simulated environment a realistic view, we performed style transfer using Low-Rank Adaptation (LoRa), where we trained a LoRa adapter that represents the realistic Thunderhill Racetrack style[15]. The training image dataset for this adapter was collected by mounting a camera for video recording on a moving vehicle on the racetrack. The vehicle drives around the racetrack for eight laps, and we sampled 3000 images uniformly from the video covering the entire racetrack. The images are then used to train a LoRa adapter that captures the aesthetic features of the racetrack.

⁴<https://github.com/chengzeyi/stable-fast>

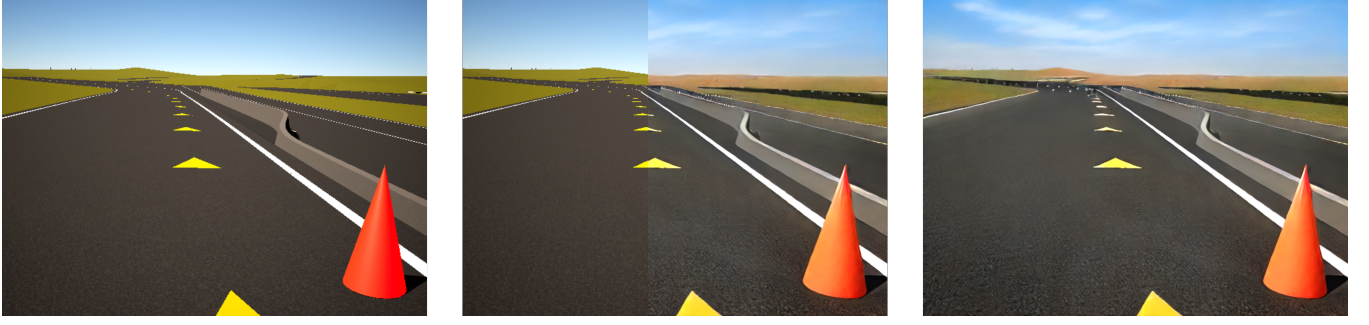


Figure 4: Simulated racetrack environment for our preliminary user study. The leftmost image is the input to the DRIVE system (condition B). The rightmost image is the output image from the DRIVE system (condition A).

4 Preliminary User Study

To assess the DRIVE system’s effectiveness in rendering realistic visuals and supporting participants in driving tasks, we run a preliminary user study where the user needs to perform driving tasks with and without the DRIVE system.

4.1 Study Setup

Our study takes place in a standard screen-based fixed-based driving simulator. A set of Logitech G29 Driving Force Racing Wheel and Floor Pedals are mounted on a rigid frame, and stationed in front of a flat monitor screen. The participants will sit in front of the steering wheel and operate the simulator as if they are in a real vehicle. For pedals, the right pedal is the acceleration pedal, the middle pedal is for the brake. The left pedal is not used in this experiment.

We chose motorsport racing as the study context because it provides a simplistic, relatively static environment, allowing participants to focus on the visual perspective of the simulation. Specifically, the participant will drive on the two-mile track counterclockwise. The racetrack is overlaid with the racing line in the form of a sequence of yellow arrows on the ground. The racing line is the optimal vehicle trajectory that would achieve the best possible lap time[2], and is collected from an expert driver’s demonstration. The participant’s task is to drive around the racetrack following these arrows for two laps. Since the frame rate is only 10 fps, we keep the maximum driving speed at 50 km/h.

4.1.1 Study Protocol. Upon arrival, the participants are asked to sign the informed consent. Then, the participants are introduced to our hardware setup. The study contains two conditions. In condition A, the participants are asked to drive two laps following the racing line in the simulated environment with the DRIVE system performing real-time style transfer. In condition B, the participants will perform the same task but see the original images without the DRIVE system enhancement. The condition order is counter-balanced. In both conditions, the simulation is running at 10 fps. Upon completion, the participants will complete a questionnaire evaluating their experience in both conditions.

4.1.2 Participants. In total, we recruited 9 participants (6 male, 3 female). All participants have a valid driver’s license, and one participant has experience with motorsport racing. The participants were

recruited based on convenience sampling within the institution, and the study has obtained IRB approval.

5 Results

5.1 Task Performance

To evaluate the users’ task performances, we first look at how well their driven trajectories match the racing line. We computed two measures: Fréchet distance and the area between trajectories [6, 17]. Fréchet distance can be thought of as a measure of maximum deviation from the racing line, and the area between curves can be interpreted as the cumulative error. While each participant drives two laps under each condition, we only consider the data collected during the second lap and treat the first lap as a warmup lap. As shown in Fig. 5 (a) and (b), both measures suggest that the users perform better under condition A (with the DRIVE system). The mean Fréchet distance is 8.23 meters with a standard deviation of 6.28. Under Condition B, the mean Fréchet distance is 11.23 with a standard deviation of 10.33. The mean cumulative area error under Condition A is 4152.6 with a standard deviation of 1128.1. The cumulative area error under Condition B is 4606.4 with a standard deviation of 1308.4.

While it seems that participants perform better under Condition A, this does not provide strong evidence that the DRIVE system improves task performance; it only demonstrates that participants are able to perform the task. As the analysis of vehicle velocity shows, participants drive slower with higher variance in speed under Condition A (mean=48.43 km/h, std=1.93) than in Condition B (mean=49.84 km/h, std=0.25). The high performance error in Condition B may be caused by participants feeling comfortable driving faster, indicating higher functional fidelity.

Therefore, from the task performance measures, we can only conclude that the DRIVE system allows participants to perform the same task as with the original simulator, without clarifying its influence on their performance.

5.2 Analysis of Questionnaire Results

Beyond numerical values collected during the task, we also analyze participants’ feedback in the questionnaire, which asks participants’ opinions about each condition in visual fidelity (how photorealistic

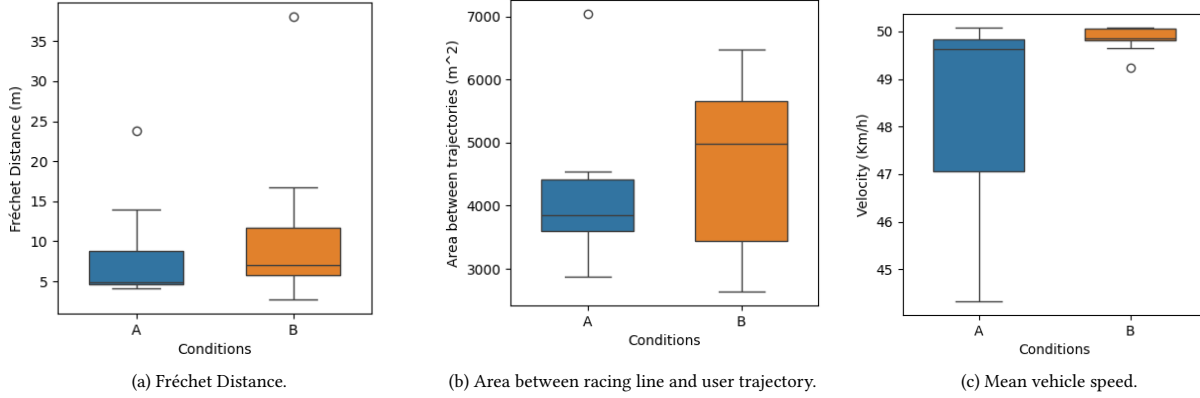


Figure 5: Subjective performance measures of the driving task.

the simulation is), and functional fidelity (how well the system supports them to perform the task in terms of maneuverability).

When we ask the participants to rate the visual realism of both conditions on a five-point Likert scale, with one being not realistic at all and five being extremely realistic, the mean rating for Condition A is 2.7, which is higher than a rating of 2.3 under Condition B. This indicates that the participants think the DRIVE system improves the visual fidelity of the simulation. Furthermore, when shown screenshots from each condition side by side, eight out of nine participants think Condition A is more photorealistic than Condition B.

We also ask the participants to rate the functional realism of both conditions on the same five-point Likert scale, which pertains to the maneuverability of the vehicle in the system and how well the system functionally supports them to complete the driving task. Both conditions have an average rating of 3.3, an indication that the DRIVE system does not improve or hinder the simulation’s original functionality in the current context.

When we ask the participants to write down their experience under Condition A (with the DRIVE system), many report that the visual fidelity is better. P1 says, “[it has] detailed sky and ground textures, smooth edges to objects.” P4 also says, “Condition B was not photorealistic compared to Condition A, as it lacked shadows or the visual details associated [with] roads, grasses, etc.” P8 reports, “[Condition A is] more realistic than B, particularly the lighting and extra detail in grass and sky.” P6, the only participant who thinks Condition B is more realistic, says “[Condition A] feels like a 2D image, so no cue of depth.” When it comes to Condition B, participants report that it is more game-like. P1 says, “[Condition B] is blocky, jagged, [and] lack of detail.” P8 comments, “The visual is flat without much texture. Looks like a basic video game.” P9 reports, “[Condition B] is less photorealistic but more high contrast.”

However, some participants report that Condition A is harder to drive within. P4 says, “While Condition A was more photorealistic, Condition B was easier to drive because the color contrast was clearer, making it easier to distinguish the objects, especially those in distant locations.” P8 also says, “Condition A was more realistic but I also got distracted that the yellow arrows were flickering between white

and yellow.” P5 comments, “It [kind of] looks like generative AI with the way each frame was coming into [the] picture. It did look more realistic. The driving seemed harder.” On the other hand, P7 says that “[in Condition B], [the] yellow marker is constant and straightforward ... I feel I am driving in a simulator but with clear instruction from the marker.” P9 reports, “The increased contrast with condition B (while not being as photorealistic) gave me more confidence in what I was seeing as a driver.” These comments may provide an explanation of why the average driving speed under Condition A is slower than that under Condition B, hinting at a slight drop in functional fidelity under Condition A.

6 Discussion

Motorsport Racing has been one of the popular contexts to collect human behavioral data [11, 14, 32, 36]. The nature of racing pushes drivers to their performance limits, yet the environment itself remains relatively straightforward compared to other driving scenarios. The simplicity of the environment, combined with clear, objective performance metrics, makes it ideal for user studies. In our research, where the focus is on enhancing visual fidelity, the racing environment provides a controlled test bed, allowing participants to concentrate on visual aesthetics without the distractions of more intricate scenarios.

6.1 Limitations

From the participants’ qualitative feedback, we notice that while the DRIVE system increases the visual fidelity of the simulation, the diffusion process may introduce factors that lead to a decrease in functional and behavioral fidelity. We discuss some of the potential reasons below and propose solutions for future works.

6.1.1 Edge Detection with Low-Resolution Input. Due to the relatively low resolution of the image feed from the simulator, the output of edge detection can be noisy, particularly for distant objects. This poses a challenge as diffusion models are highly sensitive to noise in the input control (edge) image. Even minor perturbations within the edge map can introduce undesirable artifacts, especially near the road’s vanishing point. The edge detection algorithm may

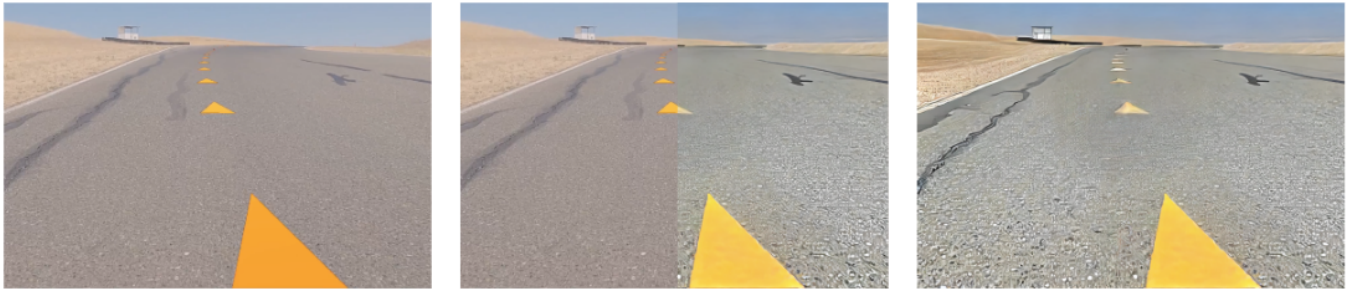


Figure 6: Image harmonization for video pass-through applications.

also struggle with complex shapes in the input images, such as clouds and signal towers. In future work, we can leverage existing world models and vehicle dynamics models to stabilize consecutive edge maps and achieve more consistent control.

6.1.2 Perception of Optical Flow. Optical flow is an important source for the perception of motion and distance [21]. In our experiment, since the race track contains long straight portions that do not induce significant scenery changes, the low-resolution simulation environment may not offer the system enough visual details (e.g. asphalt patterns) to support the perception of optical flow. As a result, this may cause the delusion that the vehicle is not moving. This is an inherent problem of the chosen simplistic environment limited by the current system, but we are optimistic that future systems can process more complex input images to support the perception of optical flow.

6.1.3 Inference Speed. The current system operates at 10 frames per second with a relatively low image resolution, suggesting considerable potential for improvement. One promising direction is the parallelization of the iterative diffusion process, as exemplified by algorithms such as StreamDiffusion and StreamV2V [20, 24]. Currently, these parallel approaches do not support the integration of additional control signals, such as those provided by ControlNet. Consequently, this approach is not adopted in our existing pipeline configuration. Nonetheless, we anticipate the future inclusion of this capability, which could significantly expedite the diffusion process.

6.2 Future Applications

Although diffusion models have not yet reached a level of maturity sufficient for deployment in safety-critical applications, our pipeline has demonstrated promising potential. Moreover, we believe our pipeline can be generalized for a variety of functionalities in the driving contexts.

6.2.1 Image Harmonization in Mixed Reality Applications. Image harmonization refers to the process of blending the foreground image with the background image [30], an approach popular among mixed reality applications. We posit that our pipeline is capable of performing real-time image harmonization in driving contexts. As shown in Fig. 6, we superimpose a virtual racing line onto video-pass-through camera footage and perform style transfer on the entire scene to achieve image harmonization.

6.2.2 Interactive Conditioning with Human Feedback. ControlNet offers an intuitive method to guide the generation process of diffusion models and allows for deliberate manipulation of the control image (edge map in our demo) based on human inputs, such as gaze or mouse cursor movements. For example, we could dynamically adjust the edge detection threshold according to the user’s gaze, refining the edge map at the focal point and coarsening it peripherally, thereby customizing spatial control in response to the user’s attention. This approach could significantly enhance interactive capabilities, providing more targeted and effective manipulation based on real-time user feedback.

7 Conclusion

In this paper, we introduce DRIVE, a system designed to enhance the realism of virtual environments using diffusion-based techniques. We outline a detailed approach to addressing the challenges of acceleration and consistency in the context of driving simulations. Our preliminary results demonstrate that while DRIVE improves visual fidelity, there remains significant room for improvement in consistency control to better capture scene details. Nevertheless, all participants were able to complete the same driving tasks with and without the DRIVE system, with the majority expressing appreciation for the enhanced visual fidelity. We encourage further development and refinement of our proof-of-concept system from the community to overcome these limitations and unlock its full potential.

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References

- [1] Gerard J Blaauw. 1982. Driving experience and task demands in simulator and instrumented car: a validation study. *Human Factors* 24, 4 (1982), 473–486.
- [2] Matteo Botta, Vincenzo Gautieri, Daniele Loiacono, and Pier Luca Lanzi. 2012. Evolving the optimal racing line in a high-end racing game. In *2012 IEEE conference on computational intelligence and games (CIG)*. IEEE, 108–115.
- [3] Yun Chen, Frieda Rong, Shivam Duggal, Shenlong Wang, Xinchun Yan, Sivabalan Manivasagam, Shangjie Xue, Ersin Yumer, and Raquel Urtasun. 2021. Geosim: Realistic video simulation via geometry-aware composition for self-driving. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 7230–7240.

- [4] Kai Cheng, Xiaoxiao Long, Kaizhi Yang, Yao Yao, Wei Yin, Yuexin Ma, Wenping Wang, and Xuejin Chen. 2024. Gaussianpro: 3d gaussian splatting with progressive propagation. In *Forty-first International Conference on Machine Learning*.
- [5] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. 2017. CARLA: An open urban driving simulator. In *Conference on robot learning*. PMLR, 1–16.
- [6] Thomas Eiter and Heikki Mannila. 1994. Computing discrete Fréchet distance. (1994).
- [7] Epic Games. [n. d.]. *Unreal Engine*. <https://www.unrealengine.com>
- [8] Martin Nicolas Everaert, Marco Bocchio, Sami Arpa, Sabine Süsstrunk, and Radhakrishna Achanta. 2023. Diffusion in style. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2251–2261.
- [9] Emilian Galazka, Teo T Niemirepo, and Jarno Vanne. 2021. CThruS2: Open-source photorealistic 3D framework for driving and traffic simulation in real time. In *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)*. IEEE, 3284–3291.
- [10] Stuart T Godley, Thomas J Triggs, and Brian N Fildes. 2002. Driving simulator validation for speed research. *Accident analysis & prevention* 34, 5 (2002), 589–600.
- [11] Marco Granato, Davide Gadia, Dario Maggiorini, and Laura A Ripamonti. 2020. An empirical study of players' emotions in VR racing games based on a dataset of physiological data. *Multimedia tools and applications* 79, 45 (2020), 33657–33686.
- [12] John K Haas. 2014. A history of the unity game engine. (2014).
- [13] Zane KJ Hartley, Rob J Lind, Michael P Pound, and Andrew P French. 2024. Domain Targeted Synthetic Plant Style Transfer using Stable Diffusion LoRA and ControlNet. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5375–5383.
- [14] Amirah 'Aisha Badrul Hisham, Marwan Nafea, Ahmad Bukhari Aujih, Mohamad Hafis Izran Ishak, and Mohamad Shukri Zainal Abidin. 2017. Racer: a simulated environment driving simulator to investigate human driving skill. In *Modeling, Design and Simulation of Systems: 17th Asia Simulation Conference, AsiaSim 2017, Melaka, Malaysia, August 27–29, 2017, Proceedings, Part II 17*. Springer, 534–547.
- [15] Edward J Hu, Yeleng Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685* (2021).
- [16] Takuya Ikeda, Suomi Tanishige, Ayako Amma, Michael Sudano, Hervé Audren, and Koichi Nishiwaki. 2022. Sim2Real instance-level style transfer for 6D pose estimation. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 3225–3232.
- [17] Charles F Jekel, Gerhard Venter, Martin P Venter, Nielen Stander, and Raphael T Haftka. 2019. Similarity measures for identifying material parameters from hysteresis loops using inverse analysis. *International Journal of Material Forming* 12 (2019), 355–378.
- [18] Nico A Kaptein, Jan Theeuwes, and Richard Van Der Horst. 1996. Driving simulator validity: Some considerations. *Transportation research record* 1550, 1 (1996), 30–36.
- [19] Seung Wook Kim, Jonah Philion, Antonio Torralba, and Sanja Fidler. 2021. Drivegan: Towards a controllable high-quality neural simulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5820–5829.
- [20] Akio Kodaira, Chenfeng Xu, Toshiki Hazama, Takanori Yoshimoto, Kohei Ohno, Shogo Mitsuhori, Soichi Sugano, Hanyang Cho, Zhijian Liu, and Kurt Keutzer. 2023. Streamdiffusion: A pipeline-level solution for real-time interactive generation. *arXiv preprint arXiv:2312.12491* (2023).
- [21] Markus Lappe, Antje Grigo, Frank Bremmer, Harald Frenz, Rene JV Bertin, and Isabelle Israël. 2000. Perception of heading and driving distance from optic flow. In *Proceedings of the Driving simulator conference*.
- [22] Kunming Li, Yu Li, Shaodi You, and Nick Barnes. 2017. Photo-realistic simulation of road scene for data-driven methods in bad weather. In *Proceedings of the IEEE International Conference on Computer Vision Workshops*. 491–500.
- [23] Shaoxu Li. 2024. DiffStyler: Diffusion-based Localized Image Style Transfer. *arXiv preprint arXiv:2403.18461* (2024).
- [24] Feng Liang, Akio Kodaira, Chenfeng Xu, Masayoshi Tomizuka, Kurt Keutzer, and Diana Marculescu. 2024. Looking Backward: Streaming Video-to-Video Translation with Feature Banks. *arXiv preprint arXiv:2405.15757* (2024).
- [25] Imanol Luengo, Evangelio Flouty, Petros Giataganas, Piyamate Wisanuvej, Jean Nehme, and Danail Stoyanov. 2018. SurReal: Enhancing surgical simulation realism using style transfer. *arXiv preprint arXiv:1811.02946* (2018).
- [26] Simian Luo, Yiqin Tan, Longbo Huang, Jian Li, and Hang Zhao. 2023. Latent consistency models: Synthesizing high-resolution images with few-step inference. *arXiv preprint arXiv:2310.04378* (2023).
- [27] Simian Luo, Yiqin Tan, Suraj Patil, Daniel Gu, Patrick von Platen, Apolinário Passos, Longbo Huang, Jian Li, and Hang Zhao. 2023. Lcm-lora: A universal stable-diffusion acceleration module. *arXiv preprint arXiv:2311.05556* (2023).
- [28] Coleman Merenda, Chihiro Suga, Joseph Gabbard, and Teruhisa Misu. 2019. Effects of vehicle simulation visual fidelity on assessing driver performance and behavior. In *2019 IEEE intelligent vehicles symposium (IV)*. IEEE, 1679–1686.
- [29] Saad Minhas, Aura Hernández-Sabaté, Shoaib Ehsan, Katherine Diaz-Chito, Ales Leonardis, Antonio M López, and Klaus D McDonald-Maier. 2016. LEE: A photorealistic virtual environment for assessing driver-vehicle interactions in self-driving mode. In *Computer Vision—ECCV 2016 Workshops: Amsterdam, The Netherlands, October 8–10 and 15–16, 2016, Proceedings, Part III 14*. Springer, 894–900.
- [30] Li Niu, Wenyan Cong, Liu Liu, Yan Hong, Bo Zhang, Jing Liang, and Liqing Zhang. 2021. Making images real again: A comprehensive survey on deep image composition. *arXiv preprint arXiv:2106.14490* (2021).
- [31] Shintaro Ono, Koichi Ogawara, Masataka Kagesawa, Hiroshi Kawasaki, Masaaki Onuki, Junichi Abeki, Toru Yano, Masami Nerio, Ken Honda, and Katsushi Ikeuchi. 2005. A photo-realistic driving simulation system for mixed-reality traffic experiment space. In *IEEE Proceedings. Intelligent Vehicles Symposium, 2005*. IEEE, 747–752.
- [32] Adrian Remonda, Eduardo Veas, and Granit Luzhnica. 2021. Comparing driving behavior of humans and autonomous driving in a professional racing simulator. *PLoS one* 16, 2 (2021), e0245320.
- [33] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
- [34] Saeed Saadatnejad, Siyuan Li, Taylor Mordan, and Alexandre Alahi. 2021. A shared representation for photorealistic driving simulators. *arXiv preprint arXiv:2112.05134* (2021).
- [35] Shital Shah, Debadepta Dey, Chris Lovett, and Ashish Kapoor. 2018. Airsim: High-fidelity visual and physical simulation for autonomous vehicles. In *Field and Service Robotics: Results of the 11th International Conference*. Springer, 621–635.
- [36] Rongkai Shi, Hai-Ning Liang, Yu Wu, Difeng Yu, and Wenge Xu. 2021. Virtual reality sickness mitigation methods: A comparative study in a racing game. *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 4, 1 (2021), 1–16.
- [37] Gaurav Sohaliya and Kapil Sharma. 2021. An Evolution of Style Transfer from Artistic to Photorealistic: A Review. In *2021 Asian Conference on Innovation in Technology (ASIANCON)*. IEEE, 1–7.
- [38] Adam Tonderski, Carl Lindström, Georg Hess, William Ljungbergh, Lennart Svensson, and Christoffer Petersson. 2024. Neurad: Neural rendering for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14895–14904.
- [39] Peter M van Leeuwen, Carla Gómez i Subils, Arnau Ramon Jimenez, Riender Happee, and Joost CF de Winter. 2015. Effects of visual fidelity on curve negotiation, gaze behaviour and simulator discomfort. *Ergonomics* 58, 8 (2015), 1347–1364.
- [40] Zhizhong Wang, Lei Zhao, and Wei Xing. 2023. Stylediffusion: Controllable disentangled style transfer via diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 7677–7689.
- [41] Zirui Wu, Tianyu Liu, Liyi Luo, Zhide Zhong, Jianteng Chen, Hongmin Xiao, Chao Hou, Haozhe Lou, Yuantao Chen, Runyi Yang, et al. 2023. Mars: An instance-aware, modular and realistic simulator for autonomous driving. In *CAAI International Conference on Artificial Intelligence*. Springer, 3–15.
- [42] Xide Xia, Tianfan Xue, Wei-sheng Lai, Zheng Sun, Abby Chang, Brian Kulis, and Jiawen Chen. 2021. Real-time localized photorealistic video style transfer. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 1089–1098.
- [43] Yunzhi Yan, Haotong Lin, Chenxu Zhou, Weijie Wang, Haiyang Sun, Kun Zhan, Xianpeng Lang, Xiaowei Zhou, and Sida Peng. 2024. Street gaussians for modeling dynamic urban scenes. *arXiv preprint arXiv:2401.01339* (2024).
- [44] Jiawei Yang, Boris Ivanovic, Or Litany, Xinhao Weng, Seung Wook Kim, Boyi Li, Tong Che, Danfei Xu, Sanja Fidler, Marco Pavone, and Yue Wang. 2023. EmerNeRF: Emergent Spatial-Temporal Scene Decomposition via Self-Supervision. *arXiv preprint arXiv:2311.02077* (2023).
- [45] Jaesun Yoo, Youngjung Uh, Sanghyuk Chun, Byeongkyu Kang, and Jung-Woo Ha. 2019. Photorealistic style transfer via wavelet transforms. In *Proceedings of the IEEE/CVF international conference on computer vision*. 9036–9045.
- [46] Ekim Yurtsever, Dongfang Yang, Ibrahim Mert Koc, and Keith A Redmill. 2022. Photorealism in driving simulations: Blending generative adversarial image synthesis with rendering. *IEEE Transactions on Intelligent Transportation Systems* 23, 12 (2022), 23114–23123.
- [47] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 3836–3847.
- [48] Yuxin Zhang, Nisha Huang, Fan Tang, Haibin Huang, Chongyang Ma, Weiming Dong, and Changsheng Xu. 2023. Inversion-based style transfer with diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10146–10156.
- [49] Xi Zhao and Wayne A Sarasua. 2018. How to use driving simulators properly: Impacts of human sensory and perceptual capabilities on visual fidelity. *Transportation research part C: emerging technologies* 93 (2018), 381–395.
- [50] Liguang Zhou, Yinglei Song, Yichao Gao, Zhou Yu, Michael Sodamin, Hongshen Liu, Liang Ma, Lian Liu, Hao Liu, Yang Liu, et al. 2023. GarchingSim: An Autonomous Driving Simulator with Photorealistic Scenes and Minimalist Workflow. In *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 4227–4232.